

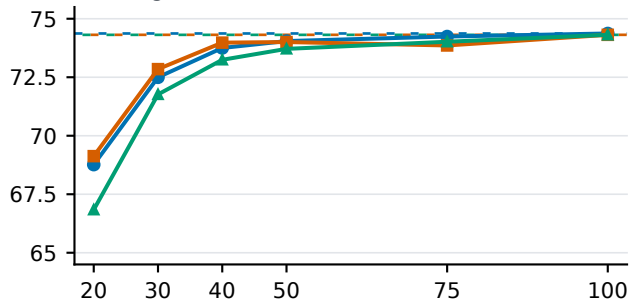
Benchmark summaries

Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

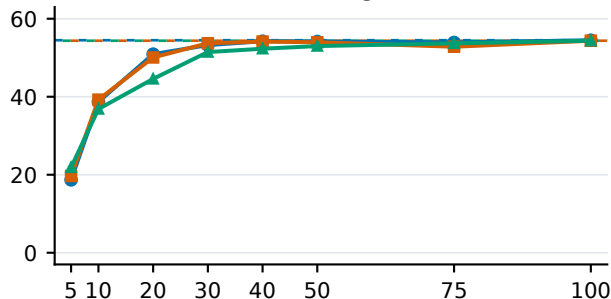
Combined evaluated suite

60 configurations; mixed benchmarks



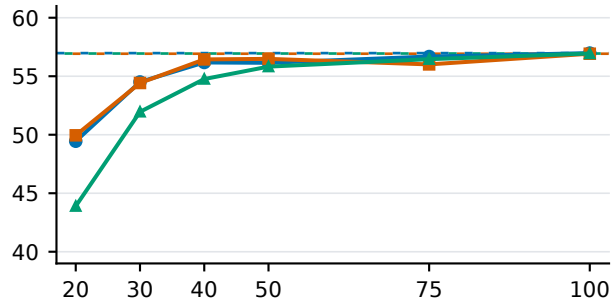
SCBench: base tasks

11 evaluated base tasks; no length variants



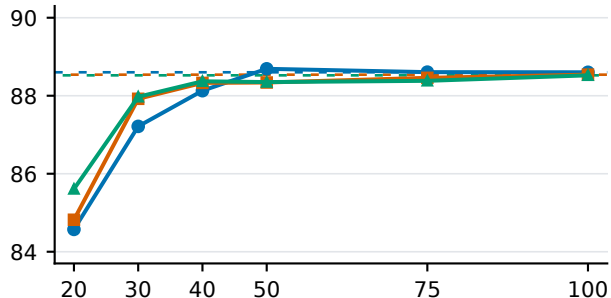
SCBench: all variants

27 configurations; variants weighted equally



RULER: evaluated lengths

33 configurations across 4K, 8K and 64K



Retention ratio (%)

Unweighted configuration means matching the appendix tables. These are summaries of the evaluated suite, not official complete-benchmark results.

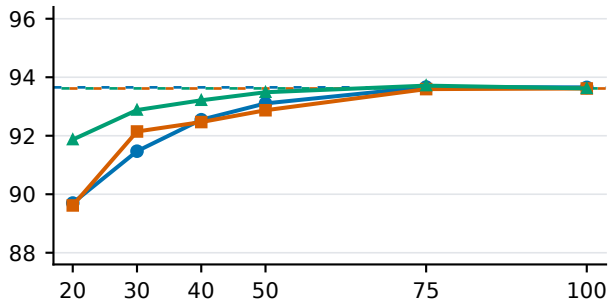
RULER summaries by context length

Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

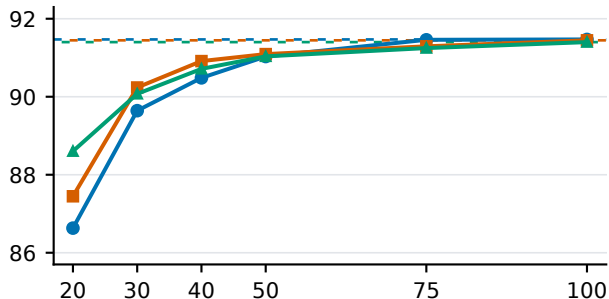
RULER 4K

13 of 13 tasks evaluated



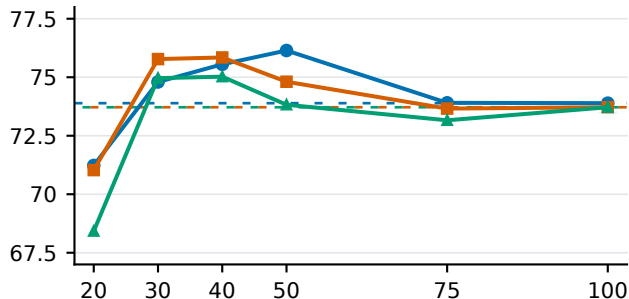
RULER 8K

13 of 13 tasks evaluated



RULER 64K

7 of 13 tasks evaluated



Retention ratio (%)

All 13 tasks are included at 4K and 8K; 64K contains seven selected tasks. Differences across lengths also reflect this unequal task coverage.

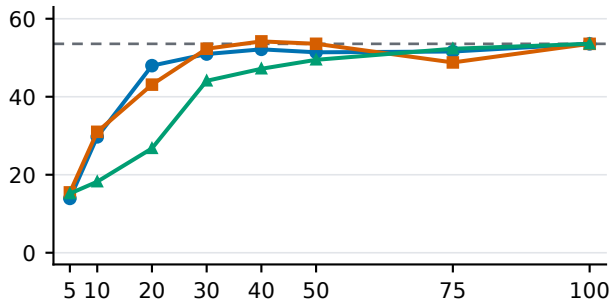
SCBench task families: base tasks

Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

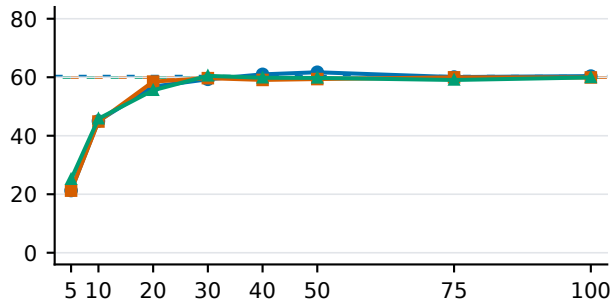
String retrieval

3 of 3 official base tasks



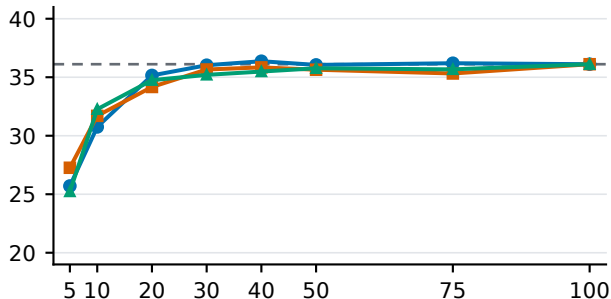
Semantic retrieval

3 of 4 official base tasks; Chinese QA absent



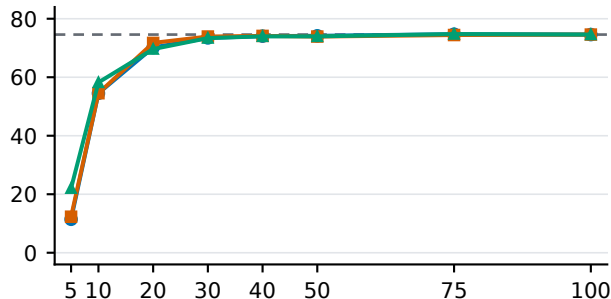
Global information processing

3 of 3 official base tasks



Multi-tasking

2 of 2 official base tasks



Retention ratio (%)

Categories follow SCBench Section 3.1 and Table 2. Semantic retrieval excludes the unevaluated Chinese-QA task. Only the 11 evaluated base tasks enter these means; prepared length variants are excluded.

SCBench task families: all variants

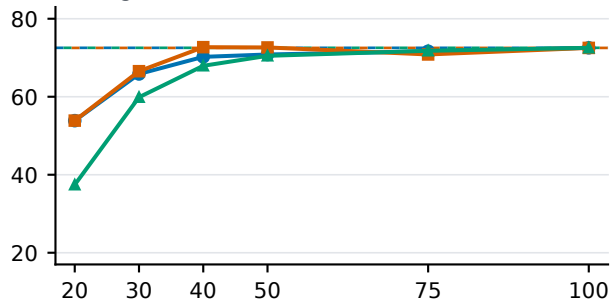
Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

String retrieval

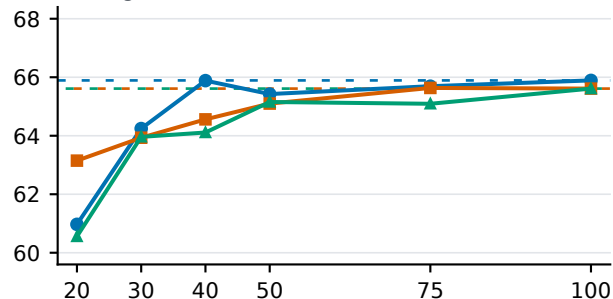
9 configurations

Score (zoomed)



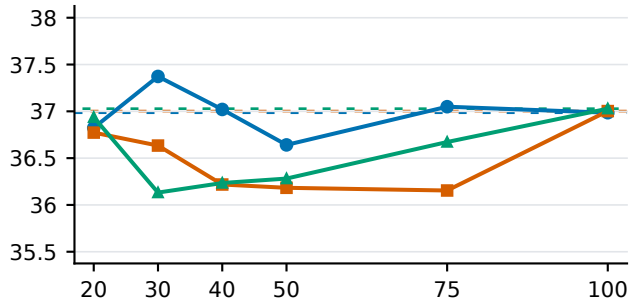
Semantic retrieval

5 configurations; Chinese QA absent



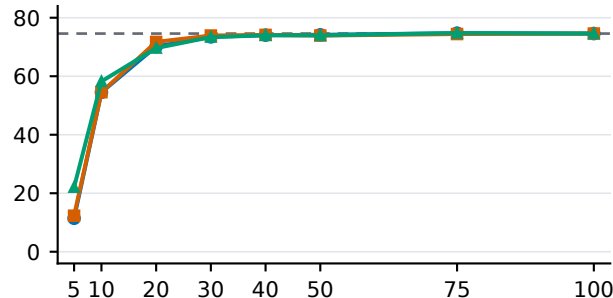
Global information processing

11 configurations



Multi-tasking

2 configurations



Retention ratio (%)

Categories follow SCBench Section 3.1 and Table 2. Semantic retrieval excludes the unevaluated Chinese-QA task. Prepared length variants inherit their base task's category and receive equal weight.

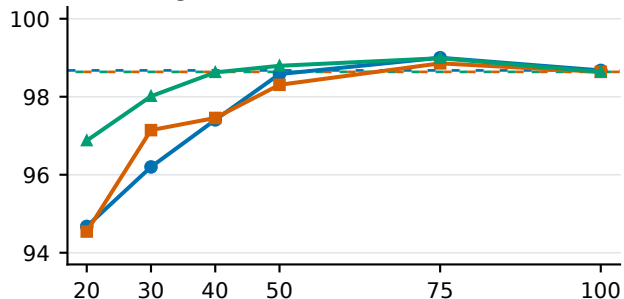
RULER 4K: task families

Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

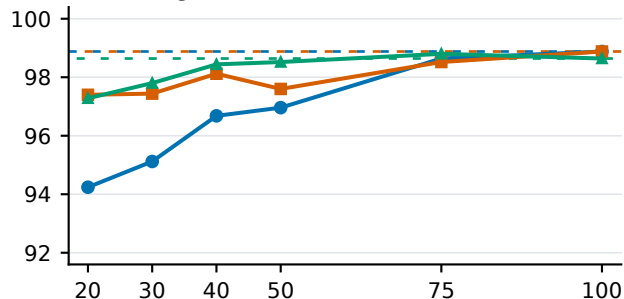
Retrieval

8 of 8 configurations evaluated



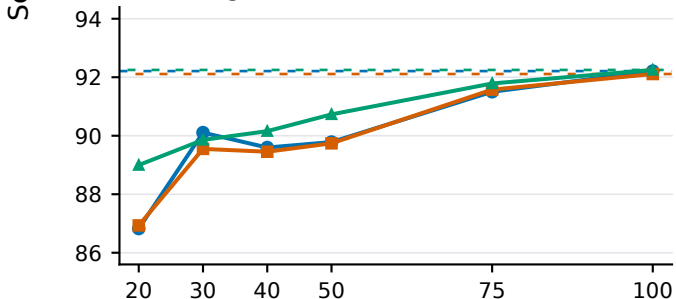
Multi-hop tracing

1 of 1 configurations evaluated



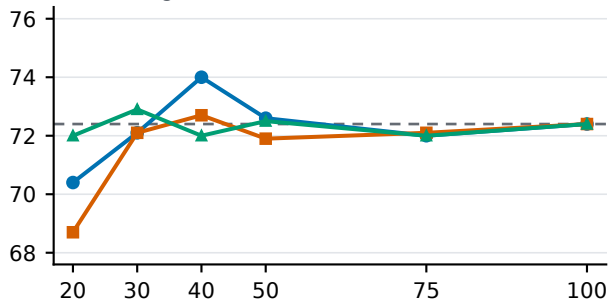
Aggregation

2 of 2 configurations evaluated



Question answering

2 of 2 configurations evaluated



Retention ratio (%)

Categories follow RULER Section 3. A one-configuration family reproduces its individual task curve.

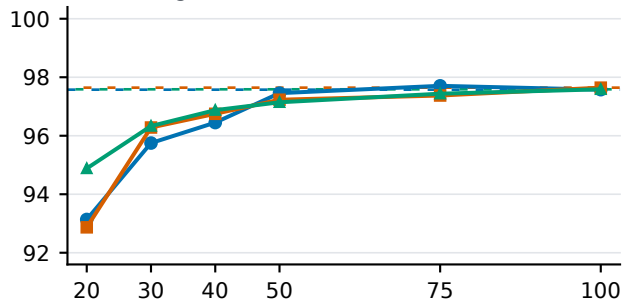
RULER 8K: task families

Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

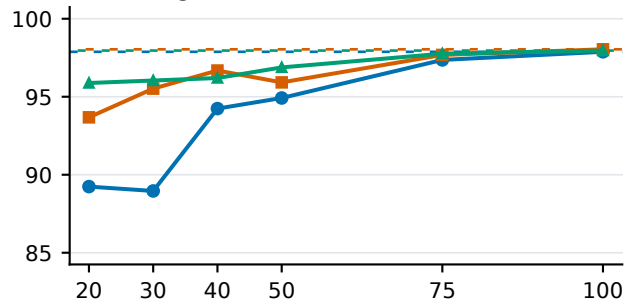
Retrieval

8 of 8 configurations evaluated



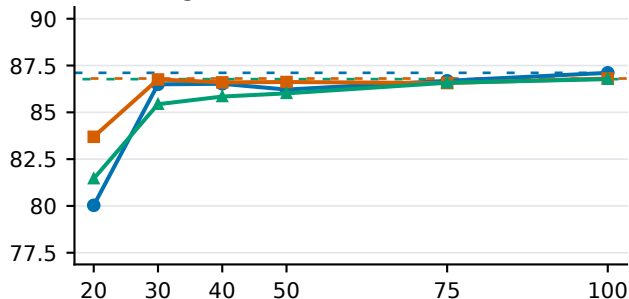
Multi-hop tracing

1 of 1 configurations evaluated



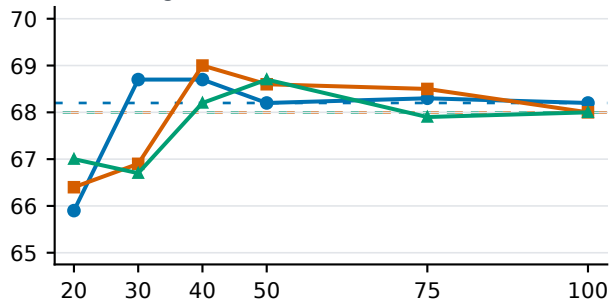
Aggregation

2 of 2 configurations evaluated



Question answering

2 of 2 configurations evaluated



Retention ratio (%)

Categories follow RULER Section 3. A one-configuration family reproduces its individual task curve.

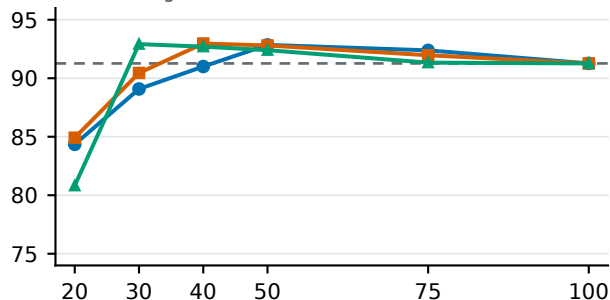
RULER 64K: task families

Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

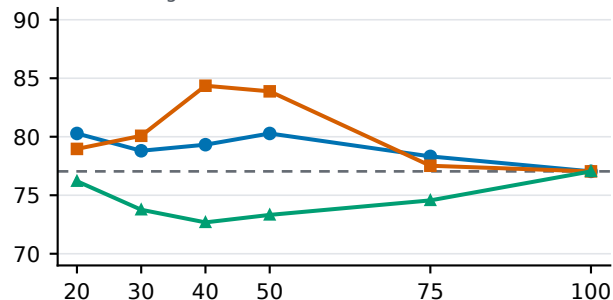
Retrieval

3 of 8 configurations evaluated



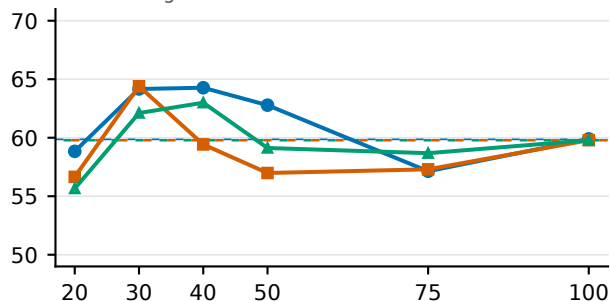
Multi-hop tracing

1 of 1 configurations evaluated



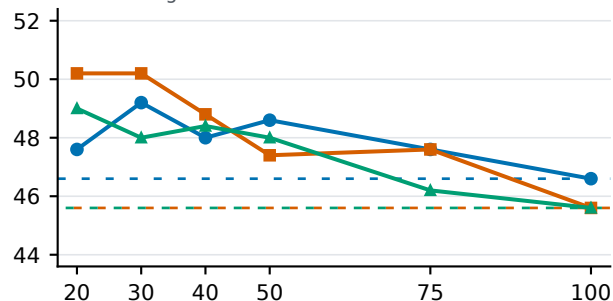
Aggregation

2 of 2 configurations evaluated



Question answering

1 of 2 configurations evaluated



Retention ratio (%)

Categories follow RULER Section 3. Only the seven evaluated 64K tasks are included. A one-configuration family reproduces its individual task curve.

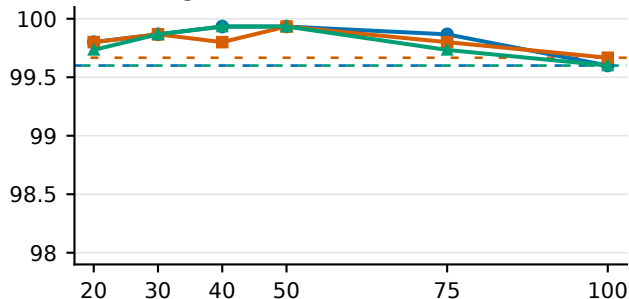
RULER 4K: retrieval subtypes

Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

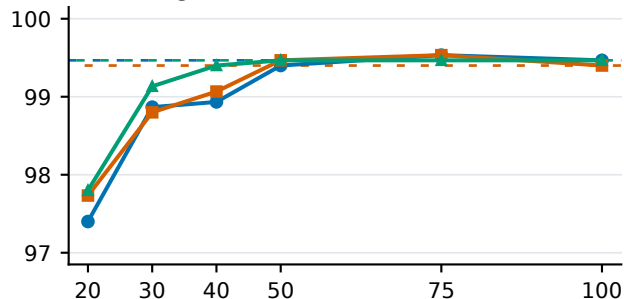
Single NIAH

3 of 3 configurations evaluated



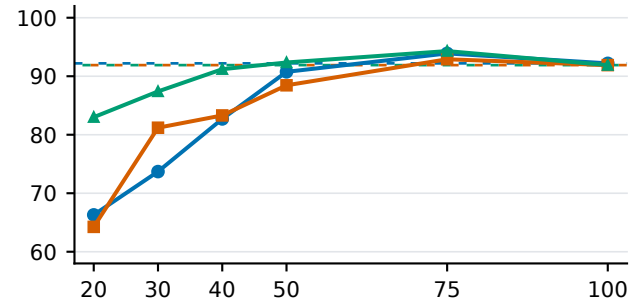
Multi-keys NIAH

3 of 3 configurations evaluated



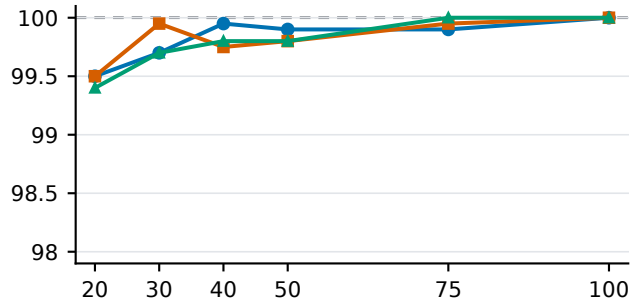
Multi-values NIAH

1 of 1 configurations evaluated



Multi-queries NIAH

1 of 1 configurations evaluated



Retention ratio (%)

Categories follow RULER Section 3. Single and multi-key means combine their available variants. A one-configuration family reproduces its individual task curve.

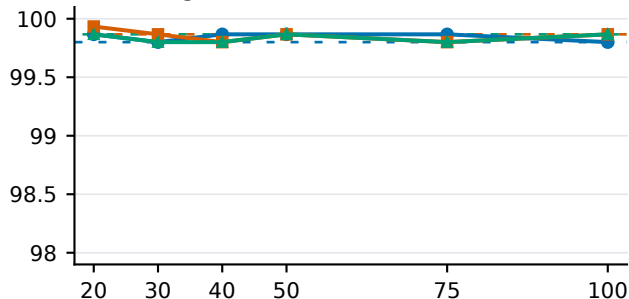
RULER 8K: retrieval subtypes

Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

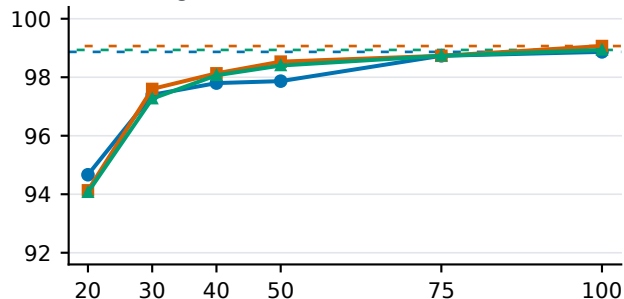
Single NIAH

3 of 3 configurations evaluated



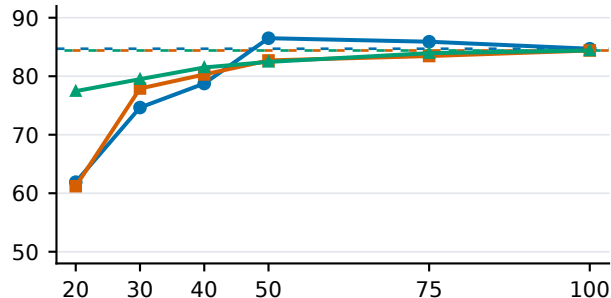
Multi-keys NIAH

3 of 3 configurations evaluated



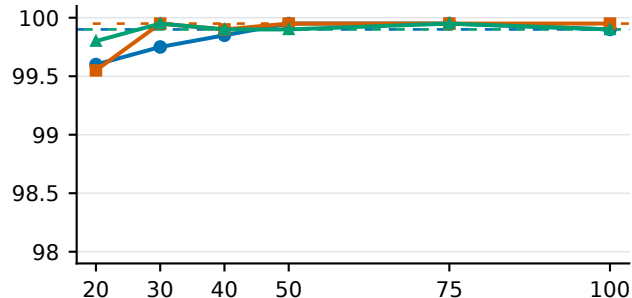
Multi-values NIAH

1 of 1 configurations evaluated



Multi-queries NIAH

1 of 1 configurations evaluated



Retention ratio (%)

Categories follow RULER Section 3. Single and multi-key means combine their available variants. A one-configuration family reproduces its individual task curve.

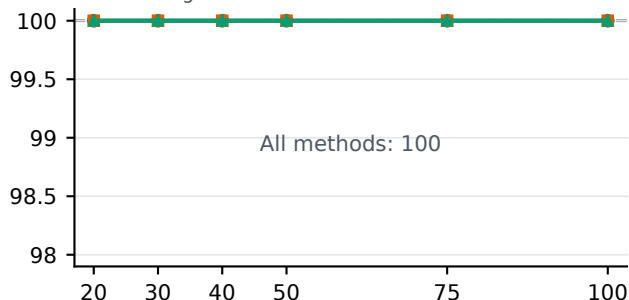
RULER 64K: retrieval subtypes

Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

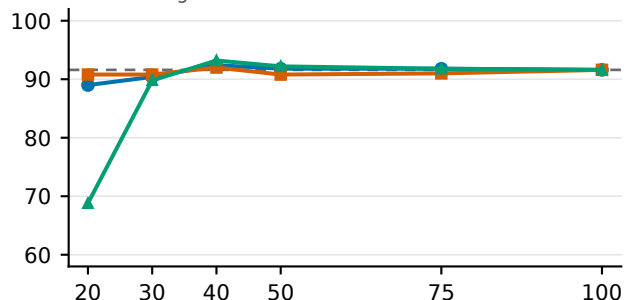
Single NIAH

1 of 3 configurations evaluated



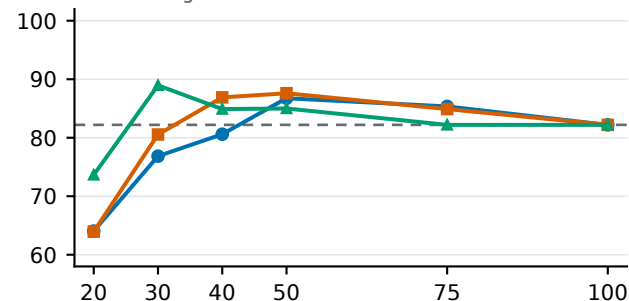
Multi-keys NIAH

1 of 3 configurations evaluated



Multi-values NIAH

1 of 1 configurations evaluated



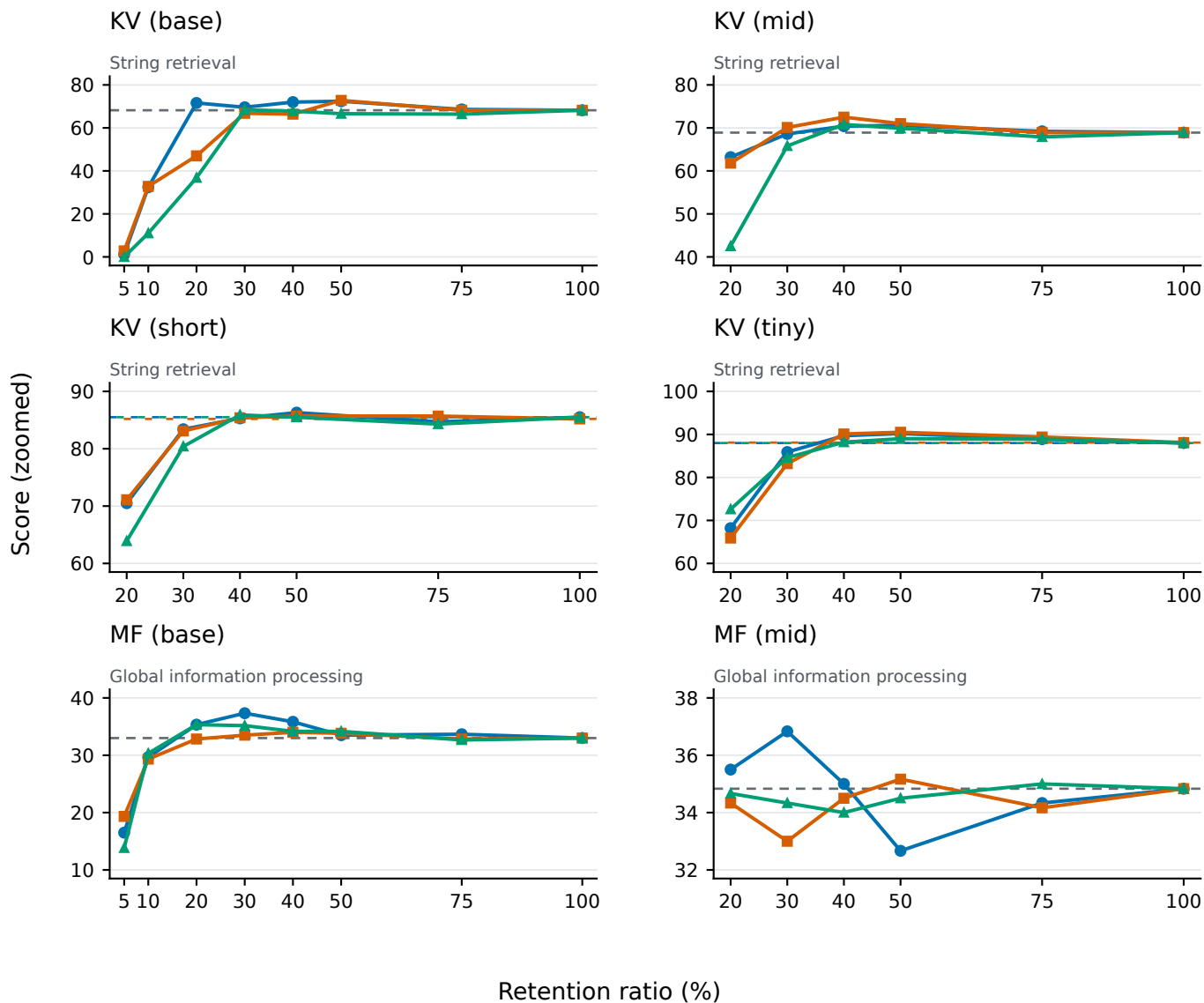
Retention ratio (%)

Categories follow RULER Section 3. Single and multi-key means combine their available variants. Multi-query retrieval is unevaluated at 64K and has no curve. Only the seven evaluated 64K tasks are included. A one-configuration family reproduces its individual task curve.

SCBench individual tasks (1/5)

Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.



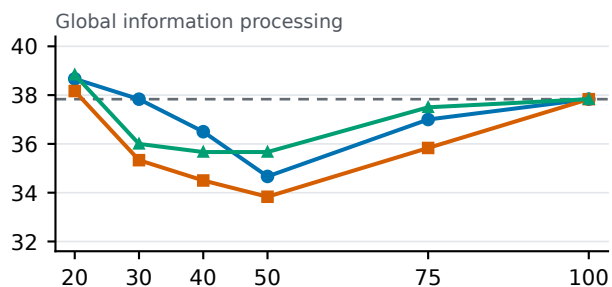
Each panel is one evaluated configuration. Base, mid, short and tiny name the prepared variants used in the tables; mixed tasks retain their deployed combined score.

SCBench individual tasks (2/5)

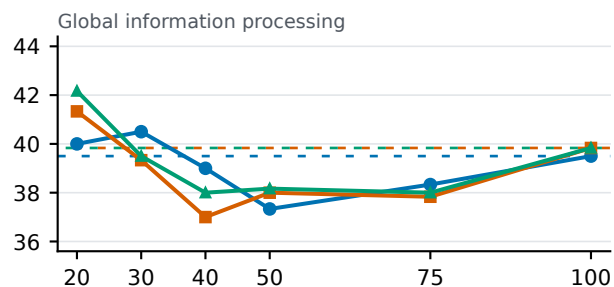
Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

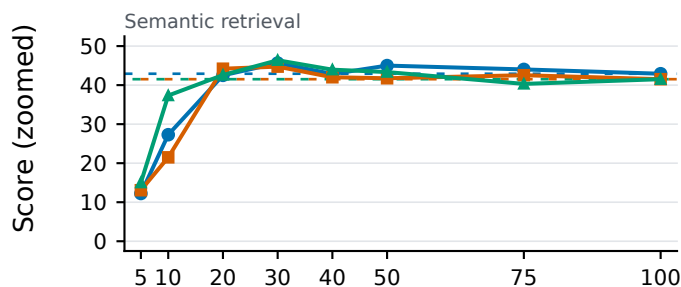
MF (short)



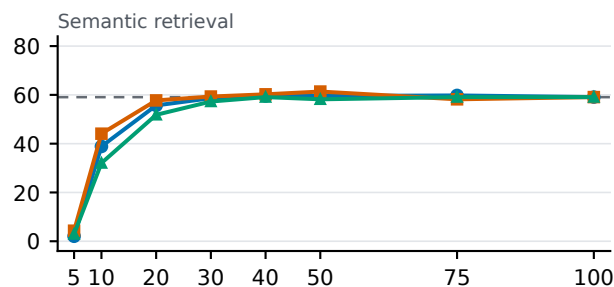
MF (tiny)



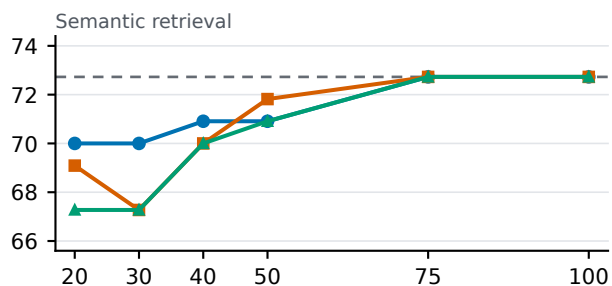
QA (English) (base)



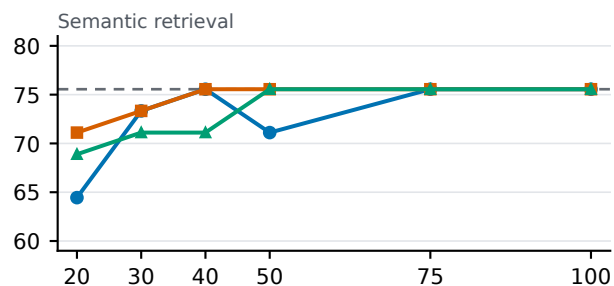
RepoQA (base)



RepoQA (short)



RepoQA (tiny)



Retention ratio (%)

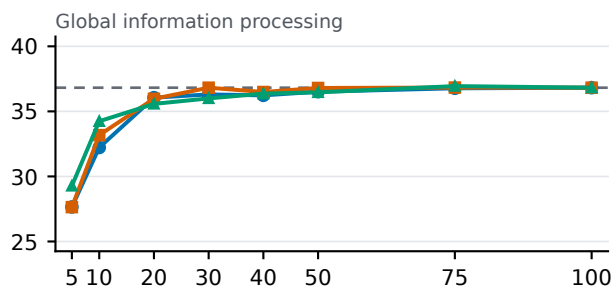
Each panel is one evaluated configuration. Base, mid, short and tiny name the prepared variants used in the tables; mixed tasks retain their deployed combined score.

SCBench individual tasks (3/5)

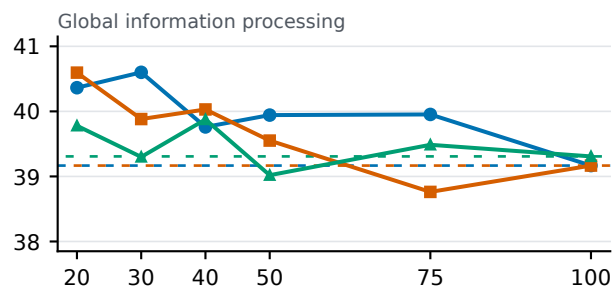
Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

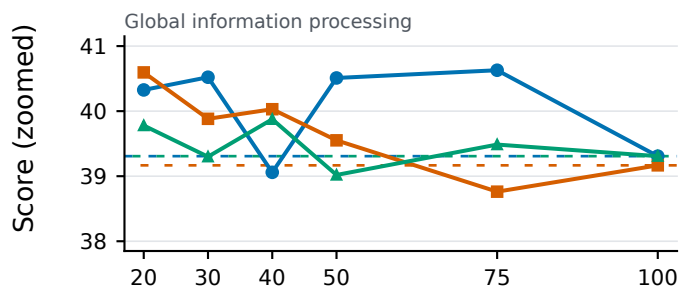
Summary (base)



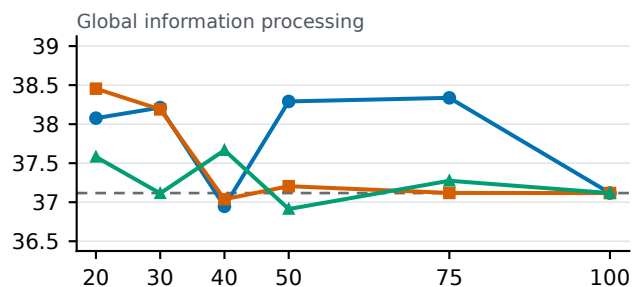
Summary (mid)



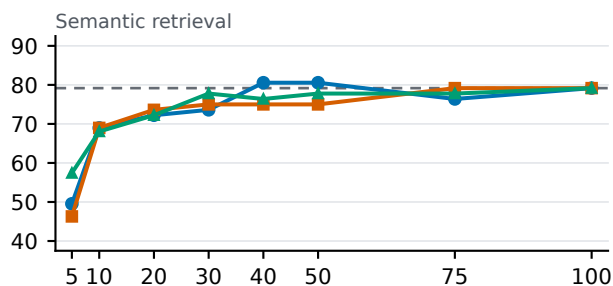
Summary (short)



Summary (tiny)



Choice (English) (base)



Retention ratio (%)

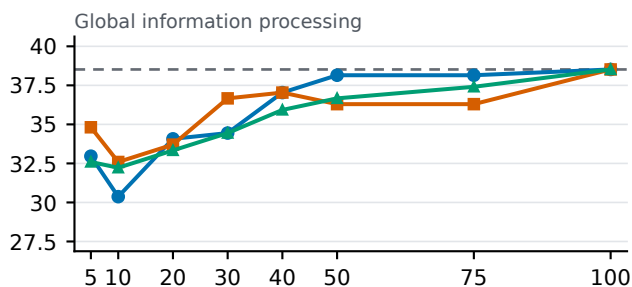
Each panel is one evaluated configuration. Base, mid, short and tiny name the prepared variants used in the tables; mixed tasks retain their deployed combined score.

SCBench individual tasks (4/5)

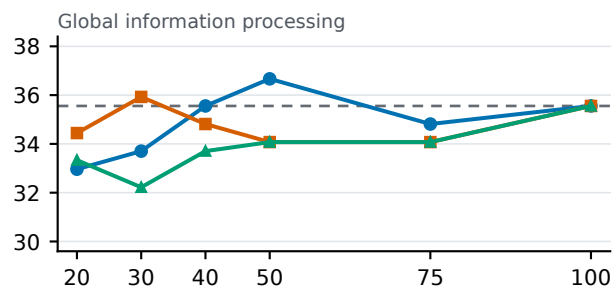
Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

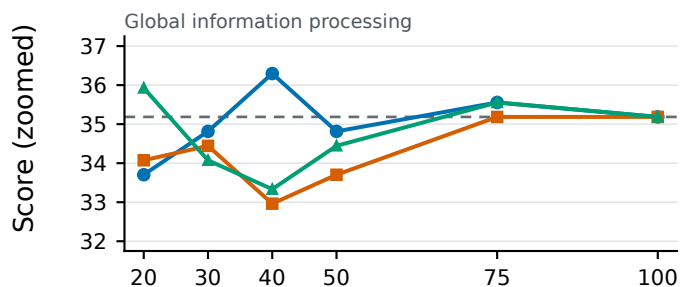
Many-shot (base)



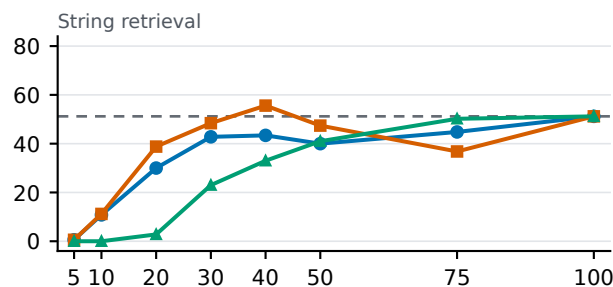
Many-shot (short)



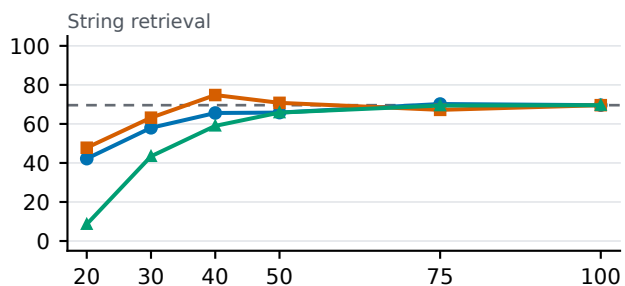
Many-shot (tiny)



Prefix/suffix (base)



Prefix/suffix (mid)



Retention ratio (%)

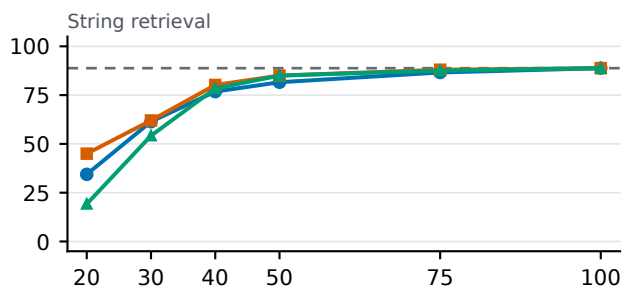
Each panel is one evaluated configuration. Base, mid, short and tiny name the prepared variants used in the tables; mixed tasks retain their deployed combined score.

SCBench individual tasks (5/5)

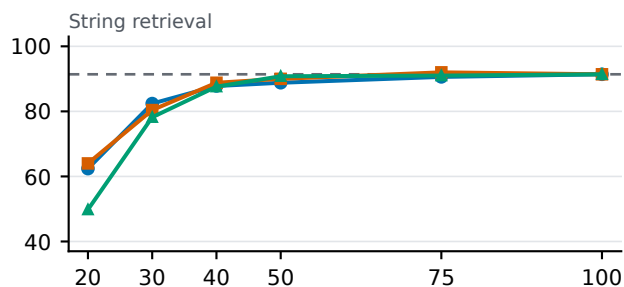
Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

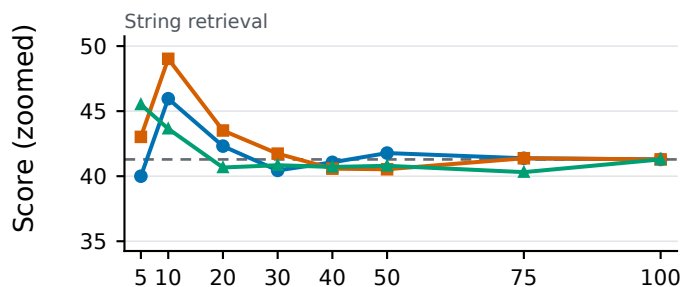
Prefix/suffix (short)



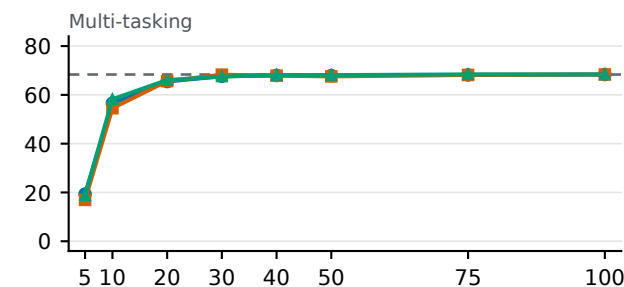
Prefix/suffix (tiny)



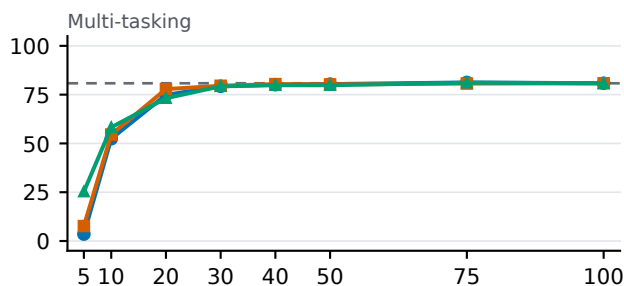
VT (base)



Summary + needles (base)



RepoQA + KV (base)



Retention ratio (%)

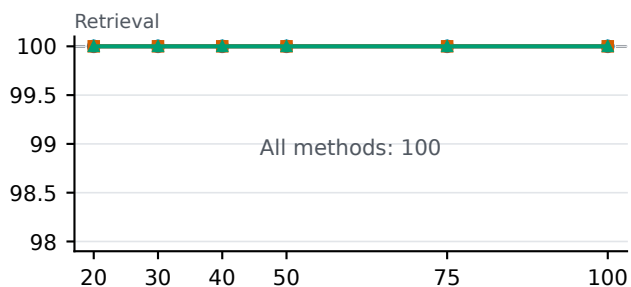
Each panel is one evaluated configuration. Base, mid, short and tiny name the prepared variants used in the tables; mixed tasks retain their deployed combined score.

RULER 4K individual tasks (1/3)

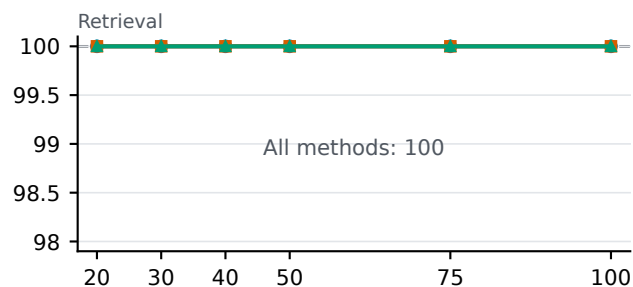
Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

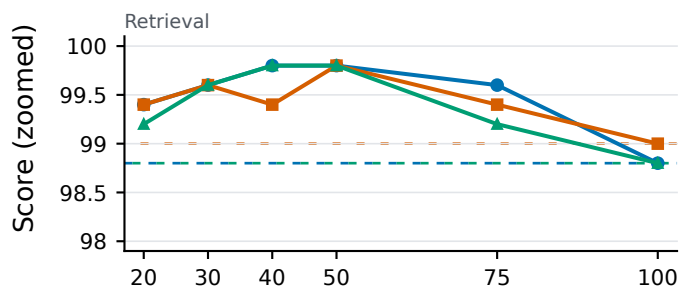
Single NIAH 1



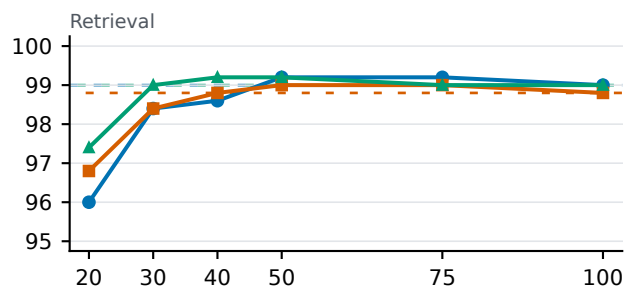
Single NIAH 2



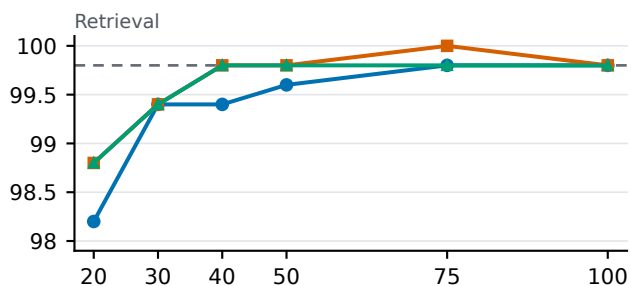
Single NIAH 3



Multi-keys NIAH 1



Multi-keys NIAH 2



Retention ratio (%)

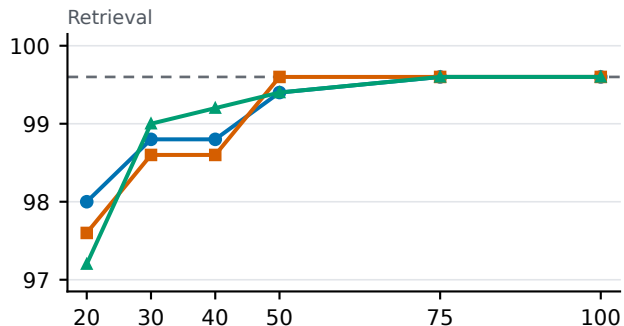
Each panel is one evaluated 4K configuration. All 13 RULER task configurations are included across this section.

RULER 4K individual tasks (2/3)

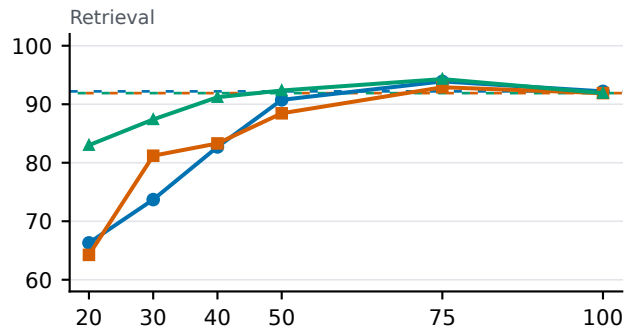
Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

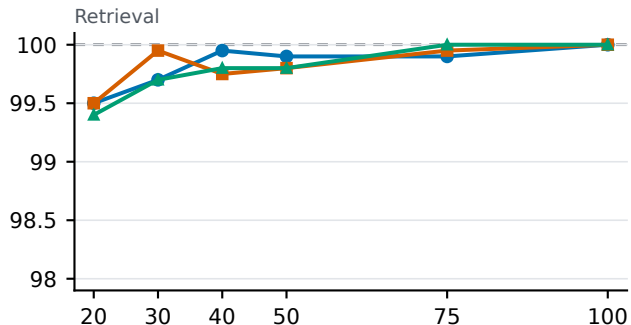
Multi-keys NIAH 3



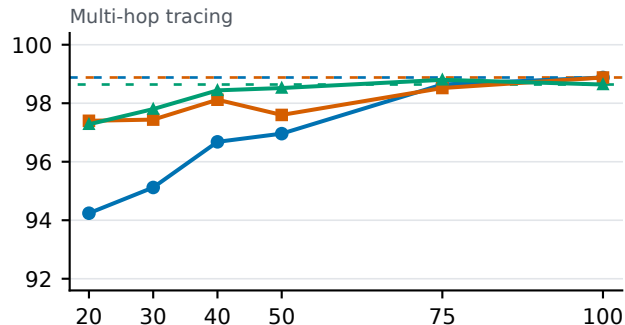
Multi-values NIAH



Multi-queries NIAH



Variable tracking



Retention ratio (%)

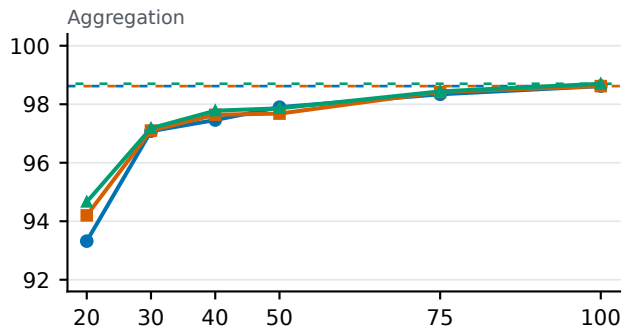
Each panel is one evaluated 4K configuration. All 13 RULER task configurations are included across this section.

RULER 4K individual tasks (3/3)

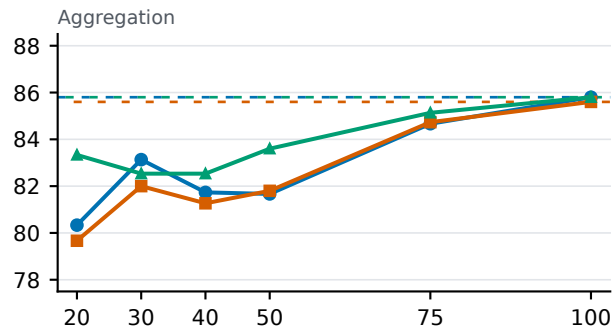
Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

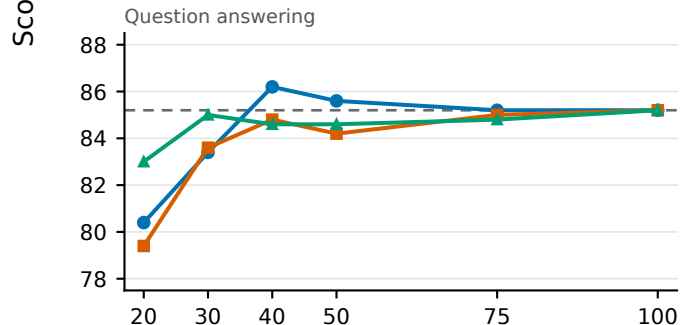
Common-word extraction



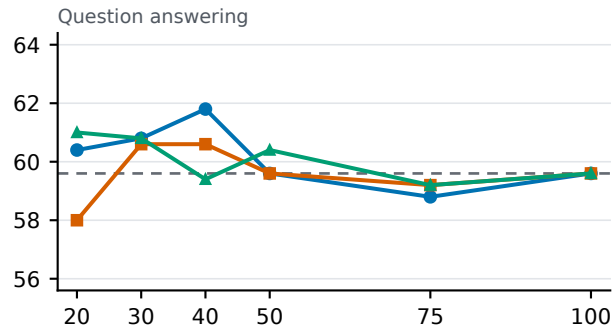
Frequent-word extraction



QA 1 (SQuAD)



QA 2 (HotpotQA)



Retention ratio (%)

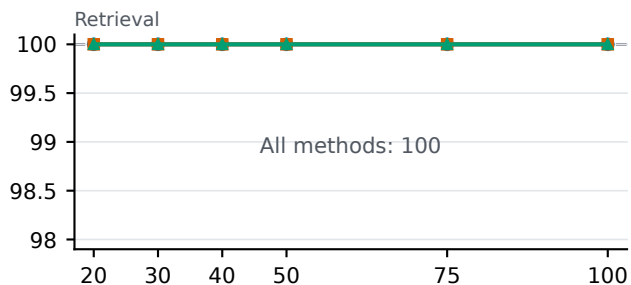
Each panel is one evaluated 4K configuration. All 13 RULER task configurations are included across this section.

RULER 8K individual tasks (1/3)

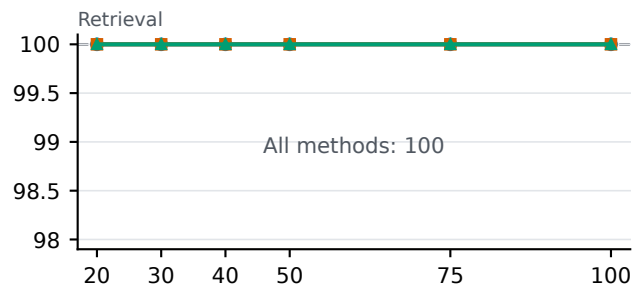
Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

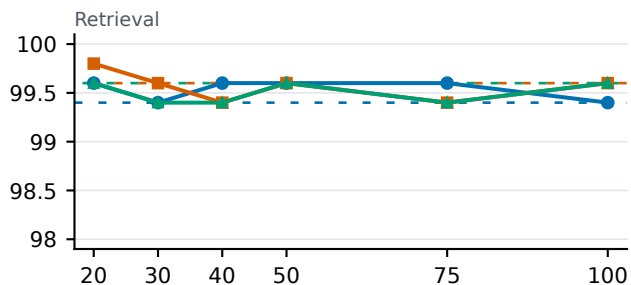
Single NIAH 1



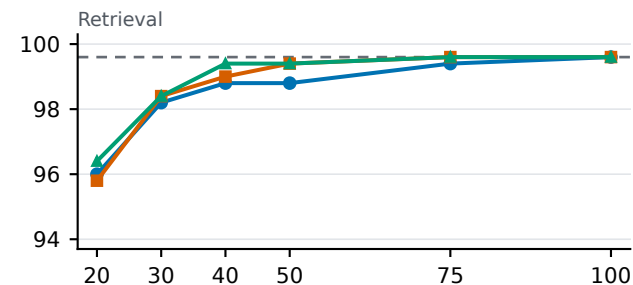
Single NIAH 2



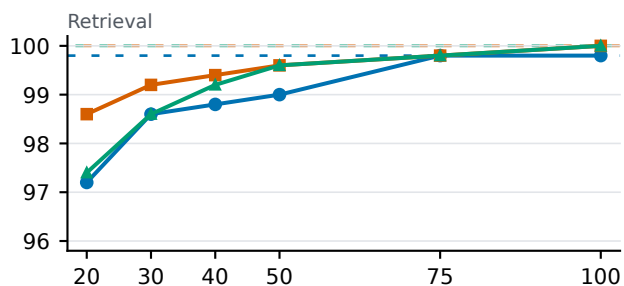
Single NIAH 3



Multi-keys NIAH 1



Multi-keys NIAH 2



Retention ratio (%)

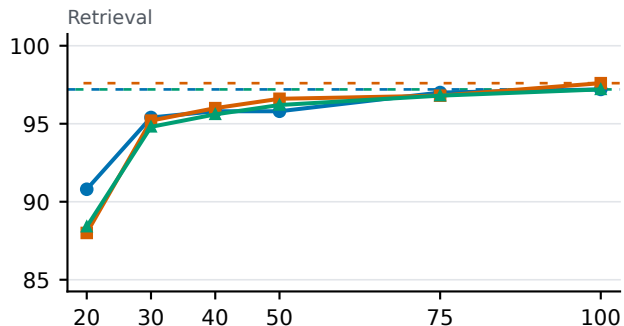
Each panel is one evaluated 8K configuration. All 13 RULER task configurations are included across this section.

RULER 8K individual tasks (2/3)

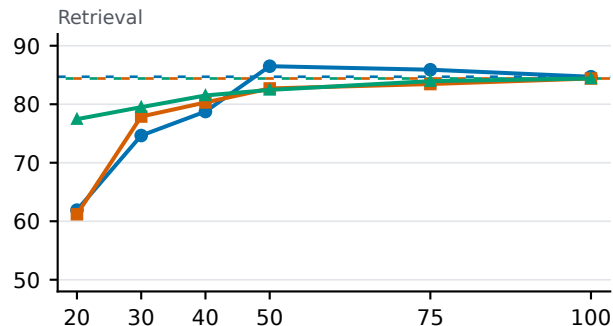
Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

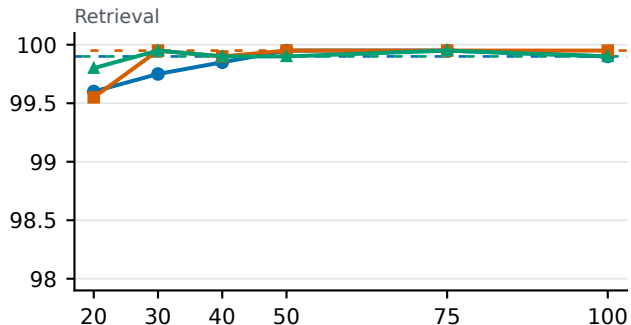
Multi-keys NIAH 3



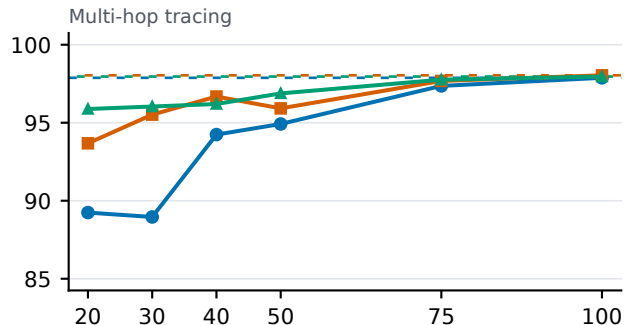
Multi-values NIAH



Multi-queries NIAH



Variable tracking



Retention ratio (%)

Each panel is one evaluated 8K configuration. All 13 RULER task configurations are included across this section.

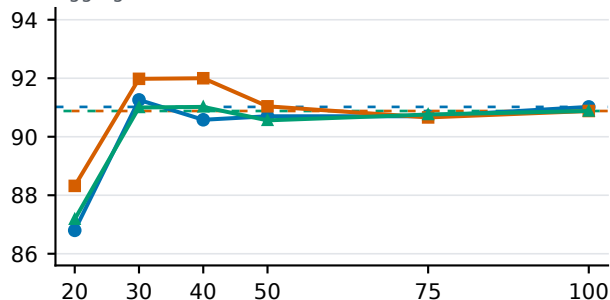
RULER 8K individual tasks (3/3)

Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

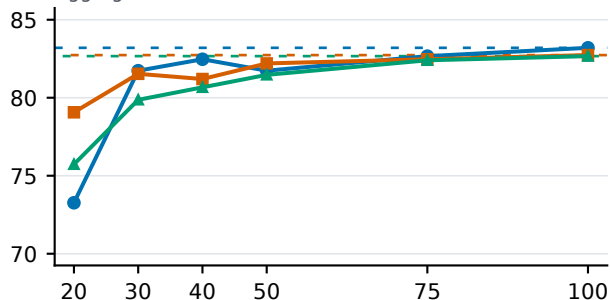
Common-word extraction

Aggregation



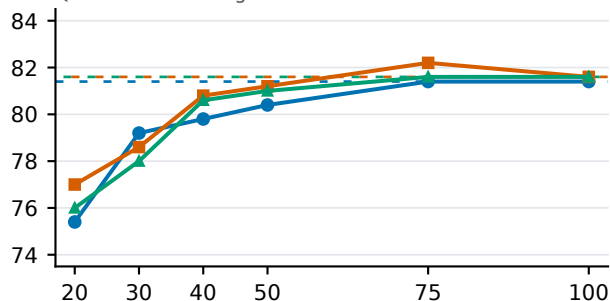
Frequent-word extraction

Aggregation



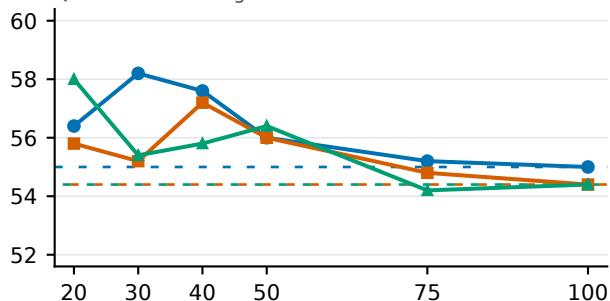
QA 1 (SQuAD)

Question answering



QA 2 (HotpotQA)

Question answering



Retention ratio (%)

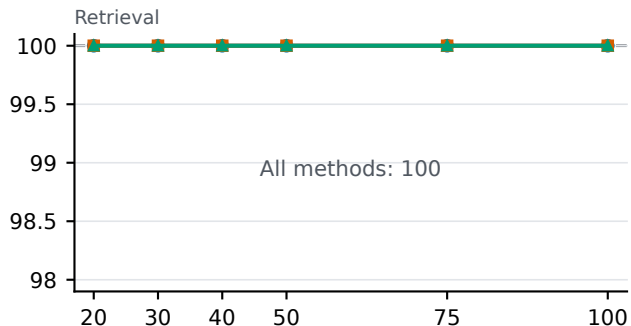
Each panel is one evaluated 8K configuration. All 13 RULER task configurations are included across this section.

RULER 64K individual tasks (1/2)

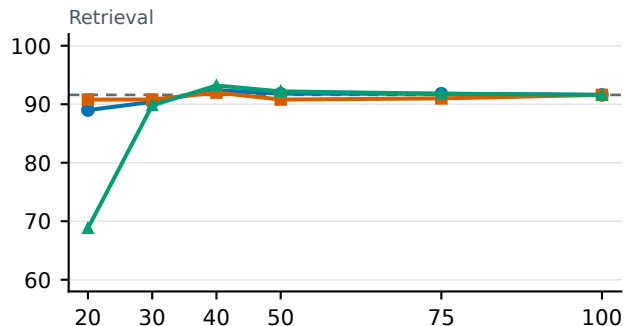
Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

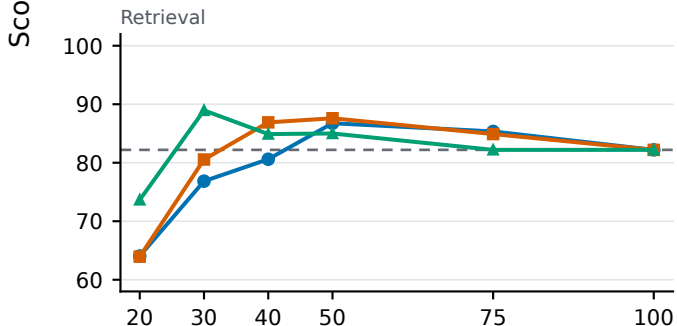
Single NIAH 1



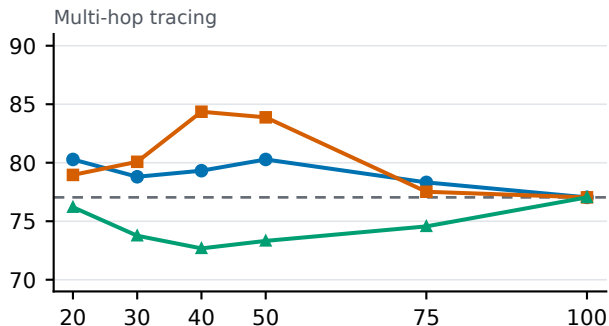
Multi-keys NIAH 3



Multi-values NIAH



Variable tracking



Retention ratio (%)

Each panel is one evaluated 64K configuration. The seven selected 64K tasks are the complete matched subset; the other six have no curves.

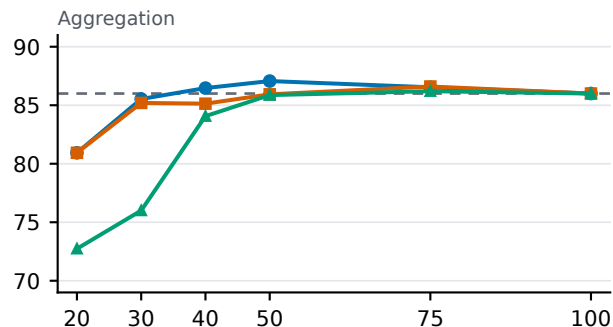
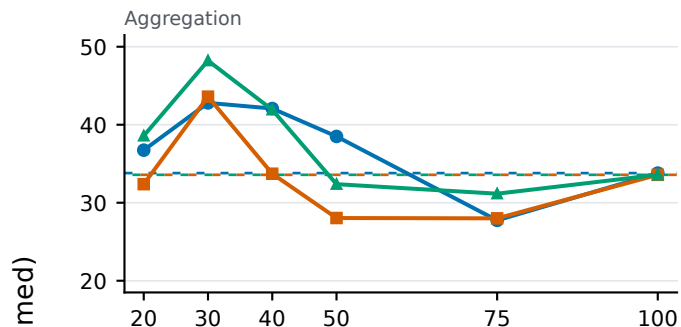
RULER 64K individual tasks (2/2)

Graph-based selector FastKVzip KVzip

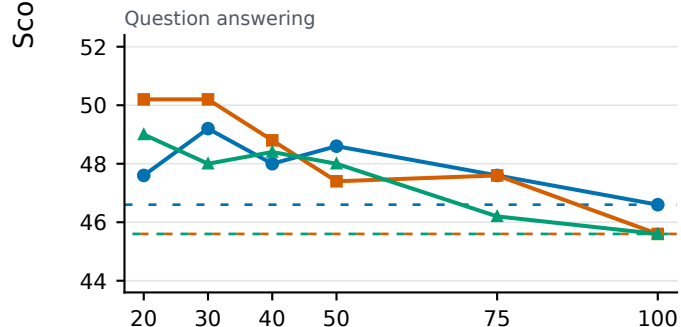
Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

Common-word extraction

Frequent-word extraction



QA 2 (HotpotQA)



Retention ratio (%)

Each panel is one evaluated 64K configuration. The seven selected 64K tasks are the complete matched subset; the other six have no curves.